

On the back-up load balancer in each data centre, run the following:

```
ipvslib12# ipvsadm --start-daemon=backup --mcast-interface=eth1
ipvslib22# ipvsadm --start-daemon=backup --mcast-interface=eth1
```

When you want to stop the daemons, you can just run the command given below:

```
# ipvsadm --stop-daemon
```

After starting the daemon on both master and backup, we can now use Heartbeat to provide high availability in our load balancers. I am not detailing the Heartbeat set-up since a similar set-up was discussed in an article that appeared earlier [*Building A Highly Available Reverse Proxy Using Heartbeat*, Pg 75, March 2009]. So this has been left as an exercise for readers. The important point here is that when the Heartbeat failover occurs, the IP address failover script sends out ARP requests to inform the nodes on the network that the VIP has been failed over and that they should update their ARP cache.

This completes the configuration. However, it would have been a lot better if we could use just one program to do all of the above, i.e., create the virtual server, monitor the virtual server and provide for automatic failover to the back-up, do the connection synchronisation, etc. There are a few tools available to do this. Some of these are *keepalived*, *ultramoney* (uses *ldirectord* and Heartbeat, and provides some add-on features), Piranha (a Red Hat favourite), etc.



Note: If you are planning to use *keepalived*, *ultramoney* or Piranha, do not execute any of the *ipvsadm* commands described above, as these applications take care of all *ipvsadm* functionalities. And if you have already executed them, it's better to give all machines a reboot to clear them.

Troubleshooting

To start with, add only one real host. In our scenario, I am assuming that you have chosen *rproxy1* (192.168.1.14) as the real host. The client IP for me was 10.1.1.1.

If you can't see the Web page, then first try pinging from the client to the VIP.

```
client# ping 192.168.1.214
```

If ping works then run the following *tcpdump* commands on the various servers:

```
director# tcpdump -n -i eth1 host 192.168.1.14
```

You should see the IPIP tunnel packets here. But if you see any ICMP error packet report that states that it could not connect, then there is a problem at the real server end. Check the tunnel there and make sure that

the link status of the *tunl0* interface is marked as UP.

```
realserver# tcpdump -n -i eth0 host 192.168.1.13
realserver# tcpdump -n -i tunl0 host 10.1.1.1
realserver# tcpdump -n -i eth0 host 10.1.1.1
```

If all seems to work well and you can see packets flowing across, then run the following *traceroute* to the client IP address on your real server. The source IP address is spoofed to be the VIP in the packets. If you cannot see the output in this command, then surely your borderline firewall or router is blocking the packets. A sample output is also shown below:

```
realserver# traceroute -n -s 192.168.1.214 10.1.1.1
traceroute to 10.1.1.1 (10.1.1.1) from 192.168.1.214: 30 hops max, 36 byte packets
 1 192.168.1.1 10.280 ms 2.700 ms 2.625 ms
 2 10.1.1.1 7.407 ms 1C 2.596 ms 1C 5.503 ms 1C
```

Try to set this up on a local LAN first before moving the set-up to the data centre scenario. And after moving to the data centre scenario, set up one data centre first. This will make troubleshooting easier.

Moving further on

In this four-part series we have seen the setting-up of various components involved in providing a highly available Web infrastructure. We have also seen how to replicate this set-up in multiple data centres and have attempted to utilise all the available capacity by balancing the traffic as evenly as possible across various components located in different data centres.

This is by no means a perfect architecture and was just an attempt to demonstrate the use of various FLOSS components in running a production infrastructure. I sincerely hope that this series has been useful to you. 

Resources

- LVS home page: www.linuxvirtualserver.org
- Ultra Monkey load balance and HA solutions: www.ultramoney.org
- Keepalived home page: www.keepalived.org
- IPVS in LVS: www.linuxvirtualserver.org/software/ipvs.html
- KTCPVS in LVS: www.linuxvirtualserver.org/software/ktcpsv/ktcpsv.html
- LVS/TUN mailing list archives: marc.info/?t=100272238300009&w=28&=1
- LVS/TUN troubleshooting: www.ssi.bg/~ja/TUN-HOWTO.txt
- Red Hat Cluster Suit configuration and management: www.redhat.net/docs/manuals/csgf/browse/rh-cs-en/index.html

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